

FOUR FOLLOW-UP QUESTIONS, ANSWERED PLAINLY

What you're actually signing up for

A companion to the main walkthrough, going deeper on the four things people ask right after: are the free services *actually* free, for how long, how does the automatic Drive lockout really work, and what is a "connector" anyway. No jargon left unexplained.

Are the free services actually free?

Yes — and it's not the kind of "free" that ends

There are two very different things people mean by "free," and it's worth separating them before anything else. A **free trial** is a countdown — thirty days, then a card gets charged. A **free tier** is a permanent allowance — you get a genuinely large amount, every month, forever, and only pay if you blow past it. All three services this project relies on are the second kind.

Think of it less like a free trial of a streaming service, and more like a mobile plan that quietly includes 750 hours of data a month, every month, for free — you'd have to use an enormous amount before you ever felt the edge of it.

SERVICE	FREE TIER LIMITS	DURATION
Render	750 instance-hours/month per workspace; sleeps after 15 min of no traffic; ~1 minute to wake back up; if you exceed 750 hours in a month, free services pause until the next month	Ongoing, no expiration — resets monthly
Turso	100 databases, 5GB storage, 500 million row reads/month, 10 million row writes/month	"Start free today, no credit card required" — ongoing
cron-job.org	Up to 60 pings/hour per job (1-minute minimum interval); no stated limit on number of jobs; no paid tier even exists	Donation-funded, free by design, no expiration

What each of those actually means day to day, in plain terms:

- **Render** is the one that actually runs the website's code. Its only real quirk: if nobody visits for 15 minutes, it goes to sleep to save resources, and the next visitor has to wait about a minute while it wakes back up — like a small shop that dims its lights when empty and needs a moment to switch everything back on when someone walks in.
- **Turso** is where all the actual data lives — every batch, task, and submission ever recorded. Its free allowance (5GB, hundreds of millions of reads a month) is enormous relative to what a submission tracker needs; a single department would need years of heavy use to get anywhere near it.
- **cron-job.org** has the simplest job of the three: it visits the website every 15 minutes just to say hello. That single visit does two things at once — wakes Render up if it was asleep, and triggers the actual Google Drive check. It's free because it's a small, donation-funded service, not because it's a limited trial.

The short version: none of this requires a credit card to set up, and none of it will silently start billing anyone later. For a tool at this scale, the free tier isn't a taste of the product — it *is* the product.

How does Drive access get revoked?

An automatic, one-time logout — with the Prof always protected

Picture the Drive submission folder as a shared office where every group has been handed a keycard. Right now, every card works. The moment the deadline plus a short grace period (24 hours, by default) passes, the system automatically deactivates every card except one — the Prof's, since the Prof owns the room.

Walking through exactly what that means:

- **When it happens** — on the next automatic check (every 15 minutes) or manual "Sync now" that occurs after the deadline plus grace period has fully passed. Not the instant the deadline hits — it waits for the next scheduled look.
- **Who loses access** — everyone currently listed with access to that folder, except the folder's owner. As a second safety net, the Prof's own connected account is protected by its email address specifically, even in the unlikely case it wasn't showing up as the actual "owner."
- **Why this needed extra permission** — simply reading files only requires "look" access to Drive. Taking away someone's key requires "manage" access, which is a bigger, more sensitive permission — this is why setting this feature up required upgrading what the app is allowed to do with the Prof's Google account.
- **If one removal fails** — it doesn't block the rest. Each person's access is revoked independently, and every one that succeeds gets written down permanently, so there's a lasting record of exactly who lost access and when.
- **It only ever happens once** — per task. Once it's succeeded, the system won't attempt it again on that same task.

LOSES ACCESS

- Every collaborator currently listed on the folder
- All at once — there's no way to selectively lock out only the late groups, since the system has no separate roster to check against

KEEPS ACCESS

- The folder's owner
- The Prof's connected Google account, matched by email as a second safeguard

The one real catch: if a deadline gets extended *after* access was already revoked, the system correctly updates everyone's On time/After labels — but it can't automatically hand back the keycards it already took. Google doesn't allow un-deleting a permission, only creating a brand-new one, so re-sharing the folder in that specific situation is a manual step for the Prof.

What is a "Google connector," concretely?

Not a Google product — just the app's own translator

A connector is the adapter plug between two things that don't naturally fit together — like a travel adapter that lets a UK plug work in a US wall socket. Google Drive speaks its own technical language; this app is written in JavaScript. The connector is the small, dedicated piece of code whose only job is translating between the two.

Concretely, in this system, it's not a separate service or a special Google feature at all — it's one file in the project, and it does exactly two things:

- **"What files are in this folder?"** — used every 15 minutes to find new or renamed submissions.
- **"Who currently has a key to this folder, and can you take one away?"** — used once, after a deadline passes, for the automatic lockout described above.

Every other part of the app that needs anything from Google Drive goes through this one piece of code — nothing else in the system talks to Google directly. That's really all "connector" means here: the single, well-defined seam between "our app" and "Google's servers," with everyone else's uploads and edits still happening in Drive exactly as they always would.

walkthrough in plain language.